Predicting Life Expectancy

Machine Learning Driven Assessment of Global Life Expectancy Trends

WHO Life Expectancy Dataset · 193 Countries · 2000 – 2015

Group J

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1 Motivation

Life expectancy is a fundamental indicator of human well-being and socio-economic development. Understanding the key factors that drive it is highly important for healthcare planning and global policy. While traditional demographic models usually rely on simple statistics, human health depends on a complex combination of non-linear factors, such as GDP, schooling, immunization rates, and nutritional levels.

In this project, we aimed to analyze these complex relationships using machine learning techniques. We used the World Health Organization (WHO) dataset, which covers 193 countries between 2000 and 2015. To meet the project requirement of using different methodologies, our group implemented three distinct algorithms:

- **Ordinary Least Squares (OLS) & LASSO Regression:** To build a parametric baseline and understand how a linear approach performs.
- **Random Forest Regressor:** To capture non-linear relationships using a tree-based ensemble (bagging) method.
- **eXtreme Gradient Boosting (XGBoost):** To achieve the highest predictive accuracy by using sequential tree boosting.

Our results show a clear improvement as we move from linear to tree-based models. The baseline LASSO model achieved a test R^2 score of 0.8195. The Random Forest model significantly improved this with an R^2 of 0.9683. Finally, our XGBoost model performed the best, reaching an R^2 of 0.9719 and a Root Mean Squared Error (RMSE) of 1.56 years. This report explains our data preparation steps, theoretical approaches, and the final evaluation of these models.

2 Data and Preprocessing

2.1 Dataset Scope

We used the longitudinal WHO Life Expectancy dataset, containing 2,938 records from 193 countries (2000–2015). The target variable is `Life expectancy` (in years). The remaining 21 features include demographic factors (`Adult Mortality`, `Infant Deaths`), economic indicators (`GDP`, `Schooling`), and healthcare metrics (`Polio`, `Diphtheria`, `Hepatitis B`).

2.2 Exploratory Data Analysis and Feature Correlations

Before implementing the machine learning models, an initial Exploratory Data Analysis (EDA) was conducted to understand the statistical distribution of the dataset. A Pearson correlation matrix was generated to identify the variables that have the strongest linear relationship with the target variable. The analysis revealed that **Schooling** and **Income Composition** act as the strongest positive predictors for a longer lifespan. Conversely, **Adult Mortality** and **HIV/AIDS** emerged as the most significant negative constraints. These correlation metrics provided a fundamental baseline, confirming that economic wealth alone (GDP) is less impactful than how that wealth is distributed (Income Composition) and invested in education.

2.3 Individual Preprocessing Pipelines

To ensure that our models are truly independent, we did not use a single pre-cleaned dataset. Instead, each group member created a custom data preprocessing pipeline that fits the specific needs of their machine learning algorithm.

2.3.1 Pipeline 1: Linear Baseline (Gulkhayo Urinova)

Linear regression models are sensitive to different scales and cannot handle missing values natively. Therefore, this pipeline focused on standardizing the data:

- **Imputation:** Missing values were filled using the global median of each feature.
- **Feature Scaling:** We applied `StandardScaler` to all numeric columns. This is mathematically necessary for LASSO regression. Without scaling, variables with large numbers (like `Measles`) would dominate the penalty function.

2.3.2 Pipeline 2: Feature-Engineered Bagging (Mert Baran Demirtas)

This pipeline focused on creating new, meaningful variables before training the Random Forest Regressor:

- **Feature Engineering:** Four new indices were created to capture socio-economic and health interactions:
  - *Dev_Index* = Income composition of resources × Schooling
  - *Immunity_Index* = (Hepatitis B + Polio + Diphtheria) / 3
  - *Health_GDP_Ratio* = Total expenditure / (GDP + 1)
  - *Avg_Thinness* = (thinness 1-19 years + thinness 5-9 years) / 2
- **Imputation:** Missing data was imputed using historical trends for each country.

2.3.3 Pipeline 3: Boosting Pipeline (Yalin Ege Dalcan)

This pipeline was designed for the XGBoost model, focusing on keeping the original data distributions intact:

- **Cleaning:** Hidden whitespaces in column names were removed using `df.columns.str.strip()` to prevent coding errors.
- **Conditional Imputation:** Instead of global averages, missing numeric values were filled by grouping the countries by their development status (`Status`):

$$\hat{x}_{i,c} = \text{median}(X_c \mid \text{Status} = \text{Status}_i) \tag{1}$$

  This ensures that a missing value for a developing country is filled with the median of other developing countries, which is statistically much more accurate.
- **Encoding and Scaling:** Categorical columns were one-hot encoded using `pd.get_dummies(drop_`. We intentionally skipped feature scaling. Since XGBoost uses tree-based threshold splits rather than distance calculations, it is scale-invariant.

3 Theoretical Background and Models

3.1 Literature Overview

Predicting life expectancy has traditionally been handled using time-series demographics. However, recent literature shows that machine learning models perform significantly better by capturing the non-linear relationships between health, education, and economy. In this project, we built upon this foundation by applying three different machine learning algorithms to compare their effectiveness.

3.2 Model 1: LASSO Regression (Baseline)

Ordinary Least Squares (OLS) regression assumes a straight-line relationship between the features and life expectancy. However, when dealing with 21 features, OLS can suffer from overfitting and multicollinearity. To prevent this, we used LASSO (Least Absolute Shrinkage and Selection Operator). LASSO optimizes the following cost function:

$$L(\beta) = \frac{1}{2n}\sum_{i=1}^{n}(y_i - x_i^T\beta)^2 + \alpha\sum_{j=1}^{p}|\beta_j| \tag{2}$$

The addition of the L_1 penalty ($\alpha\sum|\beta_j|$) mathematically shrinks the coefficients of less important features all the way to zero. This makes the model simpler and helps with automatic feature selection, extracting only the most vital predictors.

3.3 Model 2: Random Forest Regressor

To capture non-linear relationships, we moved to a tree-based ensemble approach. A single decision tree is highly prone to overfitting its training data. A Random Forest solves this by building multiple decision trees independently on different random subsets of the data (a process called bagging or Bootstrap Aggregating). The final prediction is the average of all trees:

$$\hat{y} = \frac{1}{B}\sum_{b=1}^{B} f_b(x) \tag{3}$$

This method successfully handles the complex interactions in our engineered features, such as how the combination of schooling and income affects life expectancy.

3.4 Model 3: eXtreme Gradient Boosting (XGBoost)

Our most advanced model is XGBoost. Unlike Random Forest, which builds trees independently, XGBoost builds them sequentially (boosting). Each new tree is specifically trained to correct the residual errors made by the previous trees. The objective function at step t includes both a loss metric and a regularization term Ω:

$$Obj^{(t)} = \sum_{i=1}^{n} L(y_i, \hat{y}_i^{(t-1)} + f_t(x_i)) + \Omega(f_t) \tag{4}$$

The regularization term Ω includes built-in L_1 (alpha) and L_2 (lambda) penalties to strictly control tree depth and leaf weights. For this project, we optimized the model using Grid Search to use a learning rate of 0.1, a maximum tree depth of 7, and 300 estimators.

4 Implementation

The project was implemented in Python. Below are the key snippets showing how each distinct approach was executed in code.

Snippet 1: LASSO Feature Selection (Gulkhayo)
This snippet shows the cross-validated alpha selection to find the best penalty term:

```
from sklearn.linear_model import LassoCV
# Alpha optimization with 5-fold cross validation
lasso_cv = LassoCV(alphas=[0.001, 0.01, 0.1, 1.0], cv=5)
lasso_cv.fit(X_train_scaled, y_train)
print(f"Optimal alpha selected: {lasso_cv.alpha_}")
```

Snippet 2: Feature Engineering for Random Forest (Mert)
This snippet demonstrates the creation of interaction variables before feeding them to the ensemble:

```
# Creating the Development and Immunity Indices
df['Dev_Index'] = df['Income composition of resources'] * df['Schooling']
df['Immunity_Index'] = (df['Hepatitis B'] + df['Polio'] + df['Diphtheria']) / 3
```

Snippet 3: XGBoost Hyperparameter Tuning (Yalin Ege)
This snippet highlights the grid search approach to find the most optimal boosting parameters:

```
from xgboost import XGBRegressor
from sklearn.model_selection import GridSearchCV

xgb_base = XGBRegressor(random_state=42)
param_grid = {
    'learning_rate': [0.01, 0.1, 0.2],
    'max_depth': [3, 5, 7],
    'n_estimators': [100, 200, 300]
}
grid_search = GridSearchCV(xgb_base, param_grid, cv=5,
                           scoring='neg_root_mean_squared_error')
grid_search.fit(X_train, y_train)
```

5 Training and Evaluation

5.1 Model Comparison

All models were evaluated using the same 80/20 train-test split to ensure a fair comparison. All training logs and hyperparameter configurations are fully documented within our respective Jupyter Notebooks submitted alongside this report. Table 1 summarizes the test set performances.

Student	Algorithm	Test R^2 Score	RMSE (Years)
Gulkhayo Urinova	LASSO Regression	0.8195	4.10
Mert B. Demirtas	Random Forest Regressor	0.9683	1.66
Yalin Ege Dalcan	XGBoost Regressor	**0.9719**	**1.56**

Table 1: Performance comparison of the three implemented methodologies.

5.2 Results Analysis

The evaluation metrics clearly demonstrate the advantage of tree-based ensemble methods over simple parametric models. The LASSO regression provided a solid baseline ($R^2 = 0.8195$), proving that linear correlations exist. However, it struggled with the non-linear limits of human health (e.g., healthcare spending increases life expectancy up to a certain point, after which it plateaus). The Random Forest model solved this by creating non-linear decision boundaries, dropping the error (RMSE) down to 1.66 years.

Ultimately, the XGBoost model achieved the highest predictive accuracy ($R^2 = 0.9719$, RMSE = 1.56). By continuously correcting its own errors and strictly managing variance through regularization, XGBoost proved to be the most robust architecture for this dataset.

5.3 Feature Importance and Domain Interpretation

Beyond raw predictive accuracy, our tree-based models provided crucial insights into the real-world drivers of longevity. The feature importance scores across our models consistently highlighted four critical variables:

- **Adult Mortality:** By far the strongest negative predictor. Our models prove that reducing adult death rates is the most efficient lever for increasing average life expectancy.
- **Schooling:** Access to education acts as the strongest long-term foundation for health, correlating highly with better lifestyle choices and higher income potential.
- **Income Composition:** Economic development is the engine of health. However, our models showed that it is not just about raw GDP; it is about how income is distributed and utilized within the society.
- **Disease Management (HIV/AIDS):** Specific public health threats remain significant barriers, confirming that localized health crises can offset macroeconomic gains.

6 Conclusions and Future Work

6.1 Evaluation of Results

The primary objective of this project was to predict global life expectancy by capturing the non-linear dynamics of health and socio-economic indicators. The comparative evaluation of our three machine learning pipelines clearly indicates that tree-based ensemble methods are far superior to parametric linear models for this task.

While the LASSO Regression model successfully identified the most critical predictors, it provided limited accuracy. The Random Forest model significantly reduced the prediction error by utilizing bagging techniques. Finally, our XGBoost model proved to be the most robust architecture, achieving the lowest RMSE (1.56 years) and the highest R^2 score (0.9719). If the goal is policy interpretation and simplicity, LASSO is effective. However, for high-precision healthcare planning, XGBoost is the clear winner.

6.2 Limitations and Future Steps

Although our models achieved high predictive accuracy, the research process revealed several limitations that can be addressed in future work:

- **Granular Economic Classification:** The dataset categorizes countries into a strict binary `Status` (Developed vs. Developing). This classification is overly simplistic and fails to capture the nuances of emerging economies. Future models would benefit from a continuous development index or a multi-tier classification system (e.g., High, Upper-Middle, Lower-Middle, Low Income) to better explain country-specific health outcomes.
- **Time-Series Forecasting:** In this project, we treated the data as static snapshots to predict current life expectancy, dropping the `Year` variable. A logical next step is to use time-series forecasting algorithms (such as ARIMA or LSTM neural networks) to project future values of features like GDP and Schooling, ultimately forecasting a country's life expectancy for future years.
- **Spatial and Geographic Features:** Health and economic trends often spill over national borders (e.g., regional disease outbreaks). The current dataset lacks geographical context. Incorporating spatial data, such as continent labels or geographic coordinates, could allow the model to capture regional health dynamics.

In conclusion, we successfully translated raw WHO data into a quantifiable framework for predicting, and ultimately improving, the future of human life. By understanding the statistical weights of these indicators, policymakers can allocate resources more effectively.